Jaypee Institute of Information Technology Sec-62,Noida
Department of Electronics and Communication Engineering

Micro Fabrication Technology
PBL REPORT

On

"Interactive GUI with process selection logic for thin film deposition"

Submitted by:

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ABSTRACT

Thin film deposition is a fundamental process in semiconductor fabrication and materials engineering. This project presents the development of an interactive virtual laboratory that simulates three major thin film deposition techniques: Thermal Evaporation, Sputtering, and Chemical Vapor Deposition (CVD). The simulator allows users to vary process parameters such as temperature, pressure, material type, and target thickness to observe their effects on deposition rate and required time. The system provides real-time graphical visualization of thickness growth over time, offering an intuitive understanding of process dynamics. Although simplified, the model captures essential physical relationships and serves as an effective educational tool for understanding thin film deposition mechanisms.

INTRODUCTION

Thin film deposition plays a critical role in modern electronics, particularly in the fabrication of integrated circuits, sensors, and nanostructured materials. Techniques such as Thermal Evaporation, Sputtering, and Chemical Vapor Deposition are widely used to deposit thin layers of material onto substrates.

Understanding how process parameters affect deposition behavior is essential for optimizing fabrication processes. However, industrial simulation tools are complex and not easily accessible to students. This project addresses that gap by developing a user-friendly virtual lab that demonstrates the fundamental relationships governing thin film deposition.

With the rapid advancement of microelectronics and nanotechnology, precise control over thin film properties has become increasingly critical. Even minor variations in deposition parameters can significantly impact device performance and reliability.

BACKGROUND

Thin film deposition techniques can be broadly classified into physical and chemical methods:

- **Thermal Evaporation**: A physical process where material is heated until it vaporizes and condenses onto a substrate under vacuum conditions.
- **Sputtering**: A plasma-based process where ions bombard a target material, ejecting atoms that deposit onto a substrate.
- **Chemical Vapor Deposition (CVD)**: A chemical process where gaseous reactants undergo reactions to form a solid film on a substrate.

Each method depends on different parameters:

- Temperature influences atom energy and reaction rates
- Pressure affects particle transport and collisions
- Material properties determine deposition efficiency
- Time and rate control film thickness

CVD, in particular, follows Arrhenius behavior, where deposition rate depends exponentially on temperature.

OBJECTIVE AND PROBLEM STATEMENT

Objective

To design and implement an interactive simulation tool that:

- Models thin film deposition processes
- Allows user-controlled parameter variation
- Computes deposition rate and required time
- Visualizes thickness growth using graphs

Problem Statement

Existing industrial simulation tools are complex and inaccessible for beginners. There is a need for a simplified, educational model that demonstrates how deposition parameters influence thin film growth without requiring advanced computational resources.

Expected Outcome

It should accurately demonstrate the relationship between temperature, pressure, material type, and deposition rate, along with the calculation of required time for achieving a desired thickness. The graphical visualization of thickness versus time is expected to enhance conceptual clarity, enabling users to intuitively analyze process behavior.

METHODOLOGY

The simulator models thin film deposition using simplified yet physically meaningful equations, allowing users to control key parameters such as temperature, pressure, material, and thickness. Based on these inputs, it calculates deposition rate and required time, and visualizes thickness growth through real-time graphs. This approach provides an intuitive understanding of how process parameters influence film formation.

User Inputs

The simulator allows users to control:

- Deposition Method (Evaporation, Sputtering, CVD)
- Material Type (Aluminum, Copper, Silicon, SiO_2)
- Temperature (K)
- Pressure
- Target Thickness (nm)

Mathematical Models

Thermal Evaporation:
Rate $\propto$ T/P

Sputtering:
Rate=k·T·(P+1)

(Scaled for realistic output)

CVD (Arrhenius Model)
Rate=e^−Ea/(kT)

Simulation Logic

- Deposition rate is calculated based on selected method
- Material scaling factors modify the rate
- Required time is computed as:

  $$\textbf{Time} = \text{Thickness}/\text{Rate}$$

- Thickness vs Time graph is plotted:

  $$\textbf{Thickness} = \text{Rate} \times \text{Time}$$

System Architecture

- **Frontend**: PyQt GUI
- **Backend**: Mathematical models (Python)
- **Visualization**: Matplotlib graphs

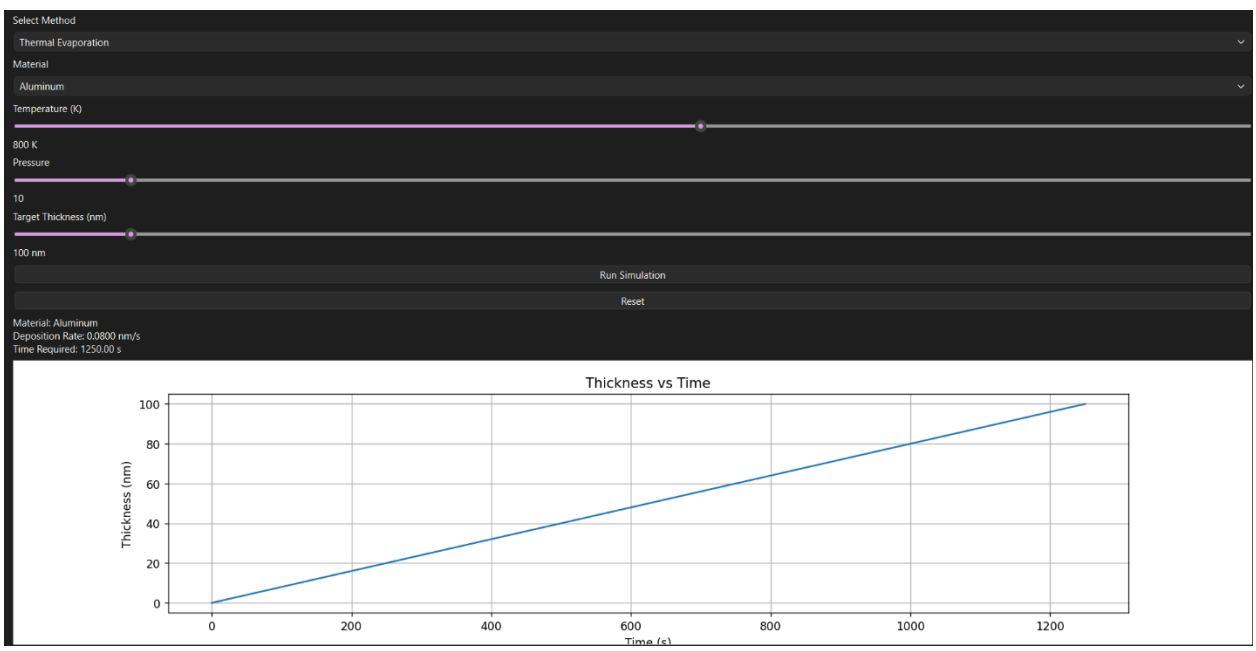
SIMULATION RESULTS

[GitHub Repository - https://github.com/quarky-1/Mircofabrication-Technology-Course-PBL]

The simulator was tested under various conditions to validate correctness.

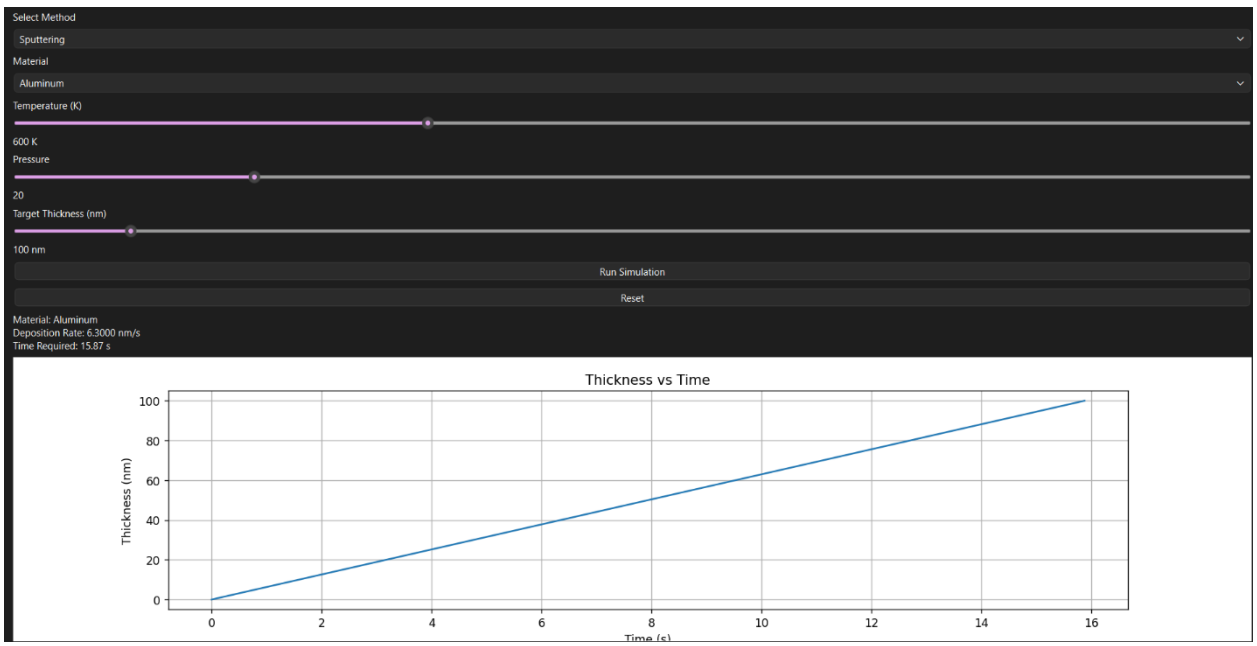
Thermal Evaporation

- Increasing temperature → higher deposition rate
- Increasing pressure → lower rate
- Linear thickness growth observed



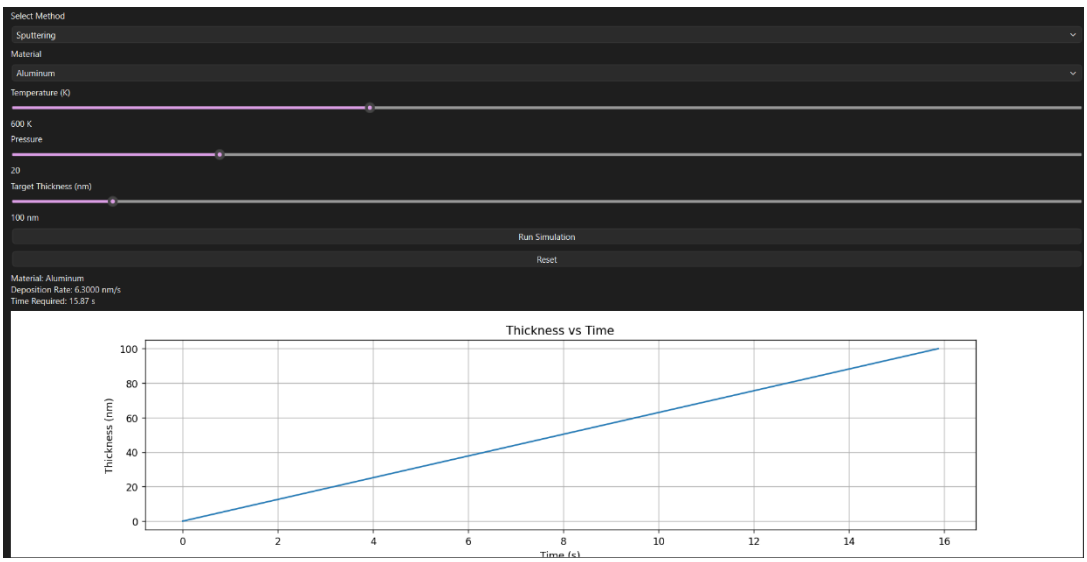
Sputtering

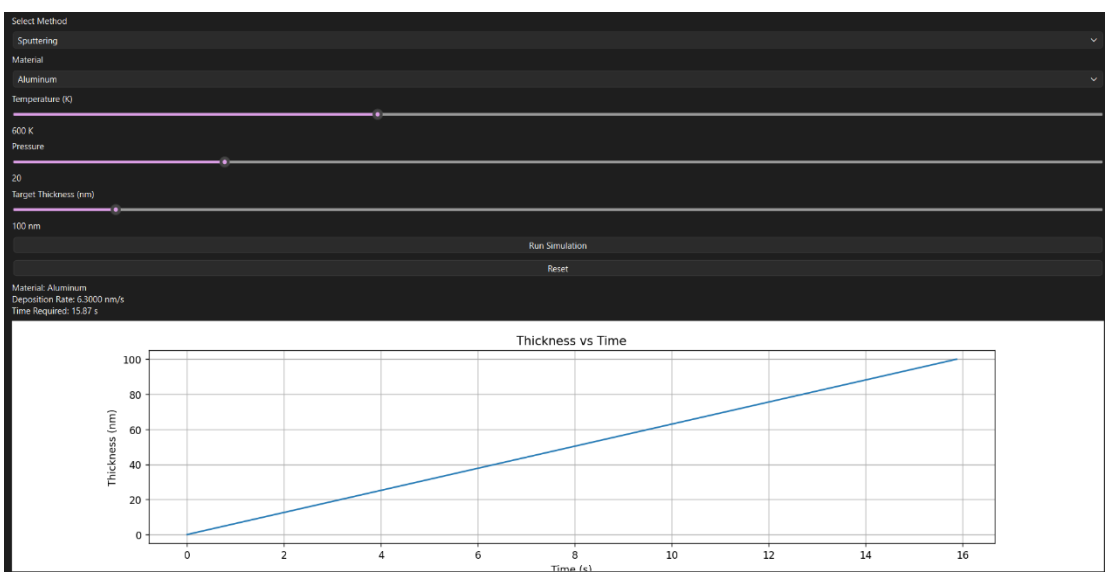
- Rate increases with both temperature and pressure
- Realistic scaling achieved after tuning constants
- Linear growth pattern confirmed



CVD

- Strong exponential dependence on temperature
- Small changes in temperature resulted in large rate changes
- Very high sensitivity observed





ANALYSIS

The simulation successfully demonstrates key physical relationships:

- **Linear Behavior**
  Thickness increases linearly with time under constant rate conditions.
- **Temperature Dependence**
  Higher temperature increases deposition rate across all methods.
- **Pressure Influence**
  - Evaporation: inverse relationship
  - Sputtering: direct relationship (model-based)
  - CVD: minimal influence in current model
- **Material Effect**
  Deposition rates vary based on material scaling factors.
- **CVD Sensitivity**
  Exponential behavior confirms Arrhenius kinetics, making it highly temperature-dependent.

Although simplified, the model preserves qualitative accuracy and reflects realistic trends.

CONCLUSION AND FUTURE SCOPE

Conclusion

This project successfully developed a virtual lab simulator for thin film deposition processes. The system provides an intuitive platform for understanding how process parameters influence deposition behavior.

Key achievements include:

- Implementation of three deposition techniques
- Accurate representation of parameter dependencies
- Interactive GUI with real-time graph visualization
- Validation through systematic testing

While the model does not replicate industrial-level simulations, it effectively captures the fundamental principles of thin film deposition and serves as a valuable educational tool.

Future Scope

- Incorporation of more accurate physical models
- Addition of multi-layer deposition simulation
- Inclusion of time-dependent rate variation
- Export of simulation data
- Integration of 2D/3D visualization.